



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

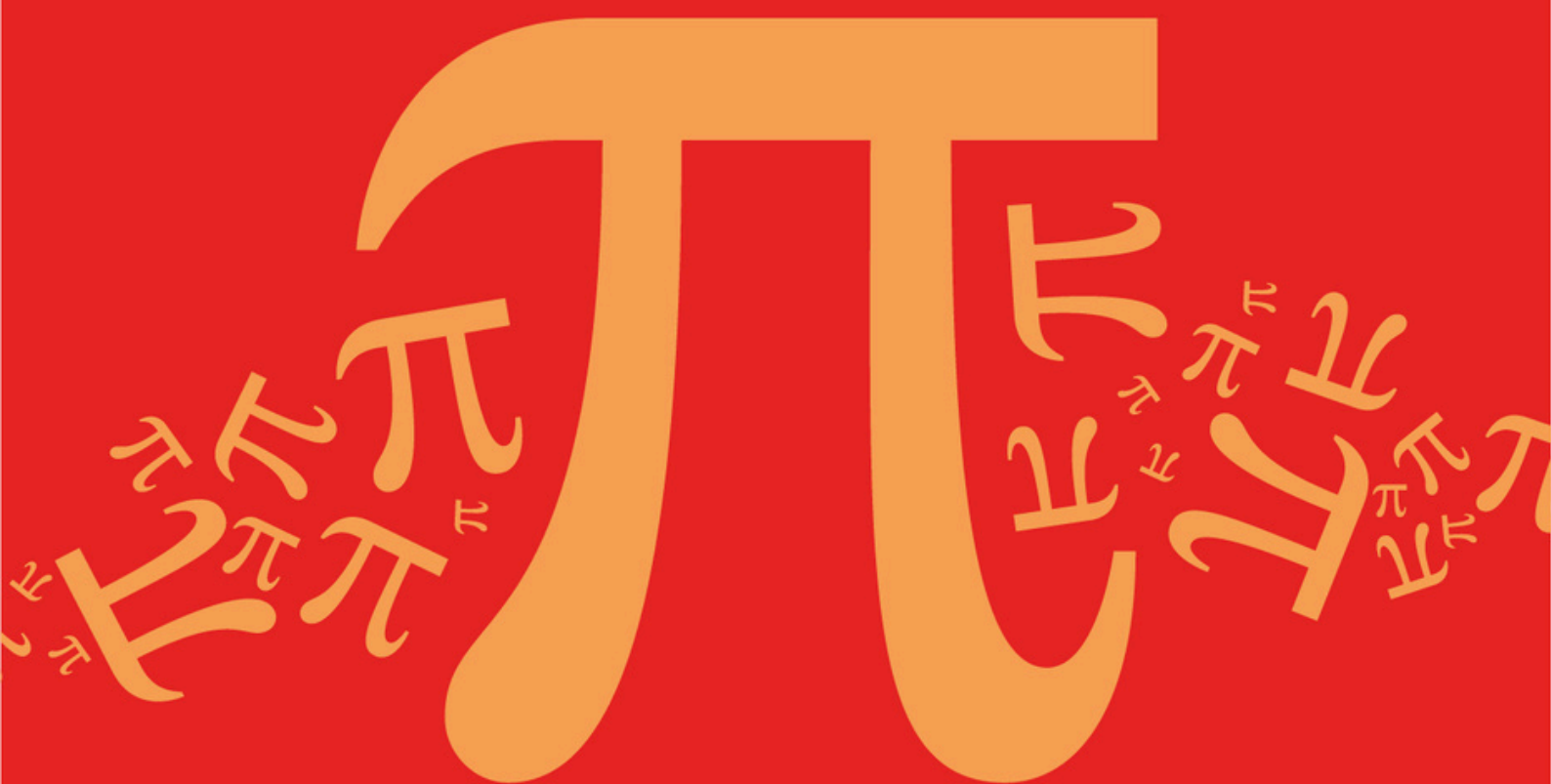
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D16: Transformations of Graphs – Rotations and Enlargements

Mark Scheme

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

(a) $\begin{bmatrix} c & s \\ -s & c \end{bmatrix}$

B1

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$$

M1

$$= \begin{bmatrix} cX + sY \\ -sX + cY \end{bmatrix}$$

A1

3

(b) (i) $(cX + sY)^2 - 6(cX + sY)(-sX + cY) - 7(-sX + cY)^2 = 8$

$$(c^2X^2 + 2csXY + s^2Y^2) - 6([c^2 - s^2]XY) + sc\{Y^2 - X^2\} - 7(s^2X^2 - 2csXY + c^2Y^2) = 8$$

Substn. for x & y in eqn. and multiplying out

M1

$$p = c^2 + 6sc - 7s^2$$

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A1

$$q = 16cs - 6(c^2 - s^2)$$

AG

A1

$$r = s^2 - 6sc - 7c^2$$

A1

(ii) Factorising:

$$3s^2 + 8sc - 3c^2 = (3s - c)(s + 3c) = 0$$

Or by double angles

M1A1

Deducing a tan value

M1

$$\tan \theta = \frac{1}{3} \quad (\theta \text{ acute})$$

A1

$$\cos \theta = \frac{3}{\sqrt{10}}, \sin \theta = \frac{1}{\sqrt{10}}$$

Both

A1

Subst^g. sensible values back for p and r

M1

$$2X^2 - 8Y^2 = 8$$

$$\frac{X^2}{2^2} - \frac{Y^2}{1^2} = 1$$

CSO

A1

A1

8

- (iii) Since C' is a hyperbola, and it is just C rotated, it follows that C is a hyperbola

EXAM PAPERS PRACTICE

E1

1

[16]

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Examiner reports

Q1.

Although there were a few excellent, complete responses to this question, in general it turned out to reflect the lack of preparedness on the part of candidates for the demands presented to them here. The three marks in part (a) were readily and easily acquired by most candidates attempting it. However, 156 candidates were unable to proceed beyond the given result of part (a), and made no attempt at the remaining parts of this question.

In part (b)(i), it was not uncommon to find that candidates were less than comfortable with squaring out their x and y terms in the new variables X and Y , and collecting them up proved to be far more problematical than should have been the case. Rather a lot of candidates seemed to think that the XY term came exclusively from the xy term, and such efforts foundered at this point.

Few candidates got to the correct equation of the hyperbola. A suitable answer to part (b)(iii) required only an awareness that the original curve had been rotated into the final hyperbola, but only a few candidates answered it without having worked their way to it. Of these, most offered spurious reasons, such as the not uncommon reference to the suggestion that the original curve was a hyperbola because this final one was an ellipse.